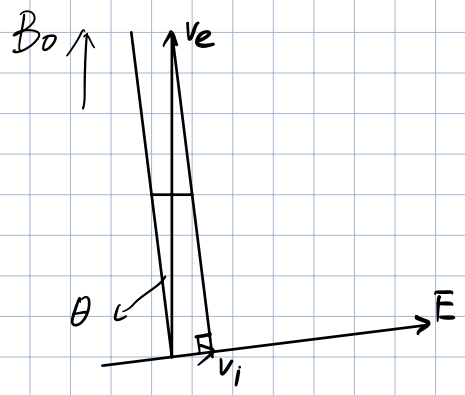
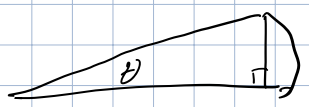


$$\frac{\pi}{2} - \theta \approx \frac{v_i}{v_e} = ?$$



$\theta \approx \sin\theta$ $\tan\theta$

$$\gamma K T_e \frac{\partial \ln n_e}{\partial z} = \frac{\gamma K T_e}{n_e} \frac{\partial n_e}{\partial z} = q E_z = q (\nabla \phi_1)_z .$$

$$\frac{\partial \ln n_e}{\partial z} = \frac{q (\nabla \phi_1)_z}{\gamma K T_e} = ik \ln n_e .$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e v_e) = 0$$

$$\frac{\partial n_e}{\partial t} + n_e \frac{\partial v_e}{\partial z} = 0 . \Leftrightarrow \underbrace{\frac{\partial \ln n_e}{\partial t}}_{-i\omega \ln n_e} + \frac{\partial v_e}{\partial z} = 0 .$$

$$\frac{\partial v_e}{\partial z} = i\omega \ln n_e = \frac{\omega}{k} \frac{q (\nabla \phi_1)_z}{\gamma K T_e}$$

$\frac{1}{2}mv^2 + e\phi$

$$f = f_0 \, e^{-\frac{\frac{1}{2}mv^2 + e\phi}{KT_e}}$$

m

$v_{//}$

$$\frac{v_e}{v_i} = \alpha \left(\frac{m}{M}\right)^{\frac{1}{2}}$$

$$\frac{v_e^2}{v_i^2} = \alpha \frac{m}{M}$$

$$M v_e^2 = \alpha m v_i^2$$

$\vec{E}$

$$\frac{\partial v_e}{\partial t} = \frac{q}{m} E_z - \frac{\gamma K T_e}{m n_e} \frac{\partial n_e}{\partial z} = \frac{e}{m} (\nabla \phi_1)_z - \frac{\gamma K T_e}{m} \frac{\partial}{\partial z} (\ln n_e)$$

$$m \frac{\partial v_e}{\partial t} = e (\nabla \phi_1)_z - \gamma K T_e \frac{\partial}{\partial z} (\ln n_e)$$

$$m \frac{\partial v_e}{\partial t^2} = e \frac{\partial}{\partial t} (\nabla \phi_1)_z - \gamma K T_e \frac{\partial}{\partial z} \frac{\partial}{\partial t} (\ln n_e)$$

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (n_e v_e) = 0$$

$$\frac{\partial n_e}{\partial t} + n_e \frac{\partial v_e}{\partial z} = 0 \iff \frac{\partial \ln n_e}{\partial t} + \frac{\partial v_e}{\partial z} = 0$$

$$\frac{\partial}{\partial z} \frac{\partial}{\partial t} (\ln n_e) + \frac{\partial^2}{\partial z^2} v_e = 0$$

$$\therefore \quad m \frac{\partial v_e}{\partial t^2} = e \frac{\partial}{\partial t} (\nabla \phi_1)_z + \gamma K T_e \frac{\partial^2}{\partial z^2} v_e$$

可能要考虑高阶项？

$$-\omega^2 m v_e = -i\omega e (\nabla \phi_1)_z - \gamma K T_e k^2 v_e$$

$$i\omega e (\nabla \phi_1)_z = (\omega^2 m - \gamma K T_e k^2) v_e$$

$$v_e = \frac{i\omega e}{\omega^2 m - \gamma K T_e k^2} (\nabla \phi_1)_z$$

$$\frac{v_e}{v_i} = \frac{\frac{i\omega e}{\omega^2 m - \gamma K T_e k^2} (\nabla \phi_1)_z}{\frac{ek}{M\omega} \phi_1 \sqrt{1 + \left(\frac{eB_0}{M\omega}\right)^2} \frac{M}{\gamma K T_e} \cdot \omega^2}$$

$$\propto \frac{1}{\omega^2 m - \underbrace{\gamma K T_e k^2}_{a}} \frac{1}{\sqrt{1 + \left(\frac{eB_0}{M\omega}\right)^2}} \qquad \text{令 } \gamma K T_e k^2 = a, \ \frac{eB_0}{\omega} = b$$

$$= \text{sqrt}^{-1}\left[(\omega^4 m^2 - 2a\omega^2 m + a^2)\left(1 + \left(\frac{b}{M}\right)^2\right) \right]$$

$$= \text{sqrt}^{-1}\left[\frac{\omega^4 m^2 b^2}{M^2} - \frac{2a\omega^2 m b^2}{M^2} + \frac{a^2 b^2}{M^2} + \omega^4 m^2 - 2a\omega^2 m + a \right]$$